



An explainable model for predicting Worsening Heart Failure based on genetic programming

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ARTICLE INFO

Keywords:

Heart failure
Worsening Heart Failure
Explainable diagnosis
Genetic programming
Artificial Intelligence
Explainable classification

ABSTRACT

Heart Failure (HF) poses a challenge for our health systems, and early detection of Worsening HF (WHF), defined as a deterioration in symptoms and clinical and instrumental signs of HF, is vital to improving prognosis. Predicting WHF in a phase that is currently undiagnosable by physicians would enable prompt treatment of such events in patients at a higher risk of WHF. Although the role of Artificial Intelligence in cardiovascular diseases is becoming part of clinical practice, especially for diagnostic and prognostic purposes, its usage is often considered not completely reliable due to the incapacity of these models to provide a valid explanation about their output results. Physicians are often reluctant to make decisions based on unjustified results and see these models as black boxes. This study aims to develop a novel diagnostic model capable of predicting WHF while also providing an easy interpretation of the outcomes. We propose a threshold-based binary classifier built on a mathematical model derived from the Genetic Programming approach. This model clearly indicates that WHF is closely linked to creatinine, sPAP, and CAD, even though the relationship of these variables and WHF is almost complex. However, the proposed mathematical model allows for providing a 3D graphical representation, which medical staff can use to better understand the clinical situation of patients. Experiments conducted using retrospectively collected data from 519 patients treated at the HF Clinic of the University Hospital of Salerno have demonstrated the effectiveness of our model, surpassing the most commonly used machine learning algorithms. Indeed, the proposed GP-based classifier achieved a 96% average score for all considered evaluation metrics and fully supported the controls of medical staff. Our solution has the potential to impact clinical practice for HF by identifying patients at high risk of WHF and facilitating more rapid diagnosis, targeted treatment, and a reduction in hospitalizations.

1. Introduction

Worsening Heart Failure (WHF) refers to worsening symptoms and signs of HF in patients with a known HF diagnosis, requiring hospitalization or an urgent visit with intra-venous therapy for HF [1]. Therefore, it requires a prior HF diagnosis, excluding events of new-onset HF. WHF is associated with an increased risk of hospitalizations [2–5], representing a significant challenge to the healthcare system, with a negative prognostic impact [6–8], as evidenced in the ESC HFA Long-Term Registry [9]. Nonetheless, WHF has variable clinical presentations according to severity, speed of deterioration, precipitating factors, comorbidities, symptoms, and clinical signs [1], making early

identification difficult. Indeed, the increase in intracardiac pressures has a crucial role in WHF, irrespective of left ventricular ejection fraction (LVEF), and precedes clear decompensation [10,11]. However, physical examination cannot correctly reveal the underlying haemodynamic alteration that results in WHF [1,11,12]. Several HF risk scores for patients with chronic HF have been developed to foretell mortality or WHF episodes [13,14]. Still, none have been validated for patients with a recent episode of WHF. Furthermore, measurements of natriuretic peptide plasma concentrations may recognize WHF patients early [15], strategies based on measurements of biochemical parameters, such as NT-proBNP levels, showed no benefits as compared with

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<https://doi.org/10.1016/j.combiomed.2024.109110>

Received 18 April 2024; Received in revised form 2 September 2024; Accepted 2 September 2024

Available online 6 September 2024

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usual care [16]. Consequently, it is crucial to recognize chronic HF patients at high risk of rehospitalization and WHFs and provide timely and targeted intervention.

Contrariwise, solutions based on Artificial Intelligence (AI) and Machine Learning (ML) can precisely analyze data trends and may reveal superior predictive ability [17–21]. Consequently, these techniques can play an essential role in the evolution of HF management by facilitating clinical decisions. One major advantage of AI/ML models is that they do not require statistical assumptions that are usually too strict for real-world data. Generally, a large arsenal of AI/ML advanced models, including Deep Learning (DL)-based, have been widely used to automate the interpretation of medical data [22,23]. However, most of such approaches rarely provide insights regarding the processes underlying the provided outputs. Consequently, physicians consider such models unreliable black boxes [24], and frequently they hesitate to make decisions relying on unjustified results. Physicians must be capable of understanding the model outputs in order to gain trust in the automatic decision-making process. The issue of interpretability is also recognized in other sectors and has gained significant importance to the extent that it has been addressed by two emerging research fields, namely Explainable Artificial Intelligence (XAI) [25] and Interpretable Machine Learning (IML) [26]. Generally, complex models tend to produce more accurate predictions than simpler models. However, comprehending the rationale behind a model's predictions becomes progressively challenging as its complexity increases. LIME (Local Interpretable Model-Agnostic Explanations) [27] and SHAP (SHapley Additive exPlanations) [28] are the most famous approaches used in this regard, also widely recognized as state-of-the-art. However, both approaches have limitations because they can only explain the behavior of the black-box around a specific instance to be predicted (i.e., local interpretation) and do not provide a global interpretation of the existing relationships between the variables involved and their corresponding targets. The reliability of an explainable model is contingent upon its ability to offer a comprehensive understanding of the system's behavior on a global scale. Furthermore, they simplify the interpretation of predictions, which can potentially lead to a significant loss in the accuracy of the model and compromise the final interpretation.

In this regard, Genetic Programming (GP) is a good candidate for XAI because it has the peculiarity of providing a mathematical formula that describes and, consequently, “explains” the relationship existing between the involved variables and the outcome, on a global scale. GP has been effectively utilized in the cardiovascular field [29], even though without providing an explainable mathematical formula, as well as in various e-health application domains for detecting other diseases, including cancer [18], diabetic foot [24], and COVID-19 [30], to name just a few.

Therefore, in this study, to enhance the prediction and prevention of WHF, we introduce a GP-based classifier designed to predict WHF using a limited number of parameters, while retaining easy and global interpretability of the results. Specifically, we utilize GP to develop a threshold-based binary classifier derived from a mathematical model that captures the relationship between these parameters and the onset of WHF within 12 months.

The main contributions of this work can be summarized as follows:

1. A mathematical formula is provided that describes the relationship between a few clinical parameters and the onset of WHF within 12 months.
2. Our model provides a straightforward and easy interpretation of the causes of WHF.
3. A 3D scatter graph allows clinicians to rapidly evaluate the health status of patients, enabling them to strategically and effectively adjust patient parameters.
4. A biochemical features selection is derived by the proposed approach.
5. Low creatinine or sPAP values seem to be associated with a lower incidence of WHF within 12 months of the first control visit, independently from history of CAD.

The rest of the paper is structured as follows: Section 2 discusses related works and outlines the main motivations of our study. Section 3 covers the fundamental concepts for proposal implementation and provides theoretical details of our approach. Section 4 presents the experiments and their corresponding results. Section 5 addresses the discussion and limitations of the presented approach. Finally, Section 6 concludes with final remarks and outlines future work.

2. Related works

Among all the parameters used to assess the presence of cardiac diseases, those derived from electrocardiograms are most commonly considered for diagnostic and prognostic purposes [31]. For this reason, thanks in part to the growth in computational power, several AI-based solutions have been investigated in the last decade, leveraging ECG information combined with other relevant features [32–34]. For instance, Newaz et al. [35] developed a decision-support system for the survival prediction of heart failure patients. They accomplished this by utilizing clinical records and laboratory tests collected from the Faisalabad Institute of Cardiology and the Allied Hospital in Faisalabad, Pakistan. In detail, they considered two sampling strategies (Chi-Square and Recursive Feature Elimination) to identify the most relevant features. Then, they employed the Random Forest algorithm to classify the survival prediction of heart failure patients by achieving a 76% average accuracy. Similarly, Hasan et al. [36] used the Recursive Feature Elimination and Minimum Redundancy Maximum Relevance (MRMR). More precisely, they employed the selected features to analyze the effectiveness of five notable ML-based algorithms, namely: Support Vector Machine, Logistic Regression, Decision Trees, Naive Bayes, and K-Nearest Neighbors. The results showed that the Decision Trees classifier achieved an 80% average accuracy for both considered selection techniques. In addition, the experimental results also proved that Ejection Fraction (EF) and Serum Creatinine are two relevant features in predicting heart failure.

However, due to the relatively low accuracy obtained, a large arsenal of enhanced AI models have been employed to make clinical analyses and diagnoses more accurate [37]. In this respect, one of the most important landmarks is DL-based solutions, which have gained promising results in distinguishing different categories of EF from TTE and ECG data [22,23]. For instance, Akerman et al. [38] employed a 3D Convolutional Neural Network (3D-CNN) to detect HF with preserved Ejection Fraction (HFpEF). To this end, they analyzed different TTE video clips related to around 6000 patients. The testing results, obtained from a dataset of 2971 patients, demonstrated good discrimination capabilities with an overall 95% AUC. Instead, Vaid et al. [39] employed a classic CNN to quantify the left and right ventricular dysfunction from ECG data. To this end, they used Natural Language Processing (NLP) to extract ventricular and systolic-related parameters from around 715 thousand ECGs. Then, they employed the mentioned CNN to classify the LVEF reduction into three categories (i.e., less than 40%, between 40% and 50%, and greater than 50%). The testing results, obtained from two validation steps, revealed an average AUC of 88% and 84% for internal and external validation, respectively.

Nevertheless, despite these studies showed satisfactory accuracy, the suggested approaches seldom offer a deeper understanding of the processes that generate the provided outputs.

To cope with this issue, recent advances have focused their attention on Explainable Artificial Intelligence. Bannister et al. [40] compared the capabilities of GP with the COX regression in diagnosing death for cardiovascular disease, non-fatal ictus, and myocardial infarction. The obtained results, derived from a dataset of 3873 symptomatic heart disease patients, demonstrated the GP's ability to achieve ratings comparable to those provided by Cox regression. Instead, Jalali et al. [41]

proposed a framework to diagnose coronary heart disease by combining the Support Vector Machine (SVM) with GP. In detail, they utilized SVM to conduct a feature selection step from the considered electrographic and biochemical features. Then, through the mathematical formula provided by GP, they faced a classification task. The obtained results, derived from a dataset of 303 patients and 54 initial features, demonstrated the effectiveness of the proposed framework by achieving a 90% average accuracy. Additionally, thanks to the mathematical formula powered by GP, they identified the 8 most relevant features for the diagnosis of heart failure, namely Obesity, Typical Chest Pain, Function Class, HTN, Age, St Elevation, Tinversion, and EF from TTE.

However, none of these works have addressed the issue of the WHF. In our work, in order to enhance prevention, we utilize the effectiveness of a GP combined to a threshold concept to establish an explainable correlation between several clinical parameters and the WHF.

3. Method

In this section, we begin by offering a concise overview of the background concepts utilized to implement the proposal, followed by the presentation of its implementation details.

3.1. Symbolic regression and genetic programming

Symbolic Regression (SR) is indeed a powerful method for uncovering mathematical relationships within multivariate datasets. It has gained popularity across various fields due to its ability to be implemented effectively using Evolutionary Algorithms (EAs) [19,42], such as Genetic Algorithms (GAs) [43]. This class of algorithms is based on Darwin's Theory of evolution, involving the creation of an initial population of individuals on a computer, which then evolves over time through the principles of selection, crossover, and mutation. A single individual, also known as chromosome, includes a number of genes, and its value can vary depending on the problem being addressed, whether numerical, binary, symbols, or characters. The number of chromosomes undergoing crossover and mutation is determined by the crossover rate and mutation rate values. The effectiveness of each individual is estimated by a fitness function. Chromosomes with higher fitness values are preferentially chosen for the next generation during the selection process. After numerous generations, the chromosome values tend to converge towards a specific value, representing the optimal solution for the considered problem.

The concept of performing SR using EAs originates from John Koza, who employed a GA in a technique known as Genetic Programming (GP) [44]. Therefore, similarly to the EAs, GP considers several individuals composing the population. Each individual corresponds to a mathematical function depicted by a tree structure. In this structure, the operands of the formula are represented as leaves, while the other nodes signify the mathematical operations (such as +, −, *, log, etc.). Fig. 1 depicts an example representing a complex function in (x, y) .

Similarly to EAs principles, GP performs the symbolic regression through the selection, crossover, and mutation operators. The performance of each individual, calculated across all entries of a given dataset comprising variables/labels pairs, is evaluated using a fitness function that quantifies the difference between the expected outcomes and the predicted ones. Then, as shown in Fig. 2(a), individuals are combined through the crossover operator, which exchanges some subtrees (genes) of two individuals (parents) to generate a new individual (offspring), which includes the best genetic information of its parents.

Also, in order to ensure genetic diversity, the mutation operation is actuated on a subset of the new individuals. As depicted in Fig. 2(b), mutation involves modifying one or more genes (segments of functions) based on a pre-defined mutation probability set by the user. Finally, such resulting individuals represent the new population, which is employed in the next cycle of the evolution (i.e., the next iteration). The process iterates until the fitness function reaches a desired value or a predefined stopping condition is met.

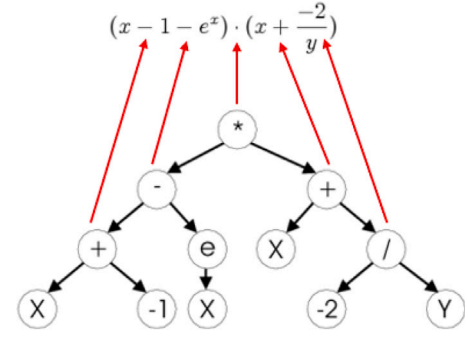


Fig. 1. A mathematical expression visualized as a tree structure.

3.2. The proposed GP-based classifier

Aim of this paper is to provide an early estimation of the WHF of a patient, and to discover the symbolic relationship (i.e., the mathematical formula) between bio-clinical parameters and WHF.

Accordingly, let R^n be the generic set of input features and let Y be the set of corresponding output classes, the goal of the GP-based classifier is to provide an interpretable formula $f^* : R^n \rightarrow Y$, which better fits the input/output pairs of a training dataset D . Notice that $D = (R^n \times Y)$, that is any item of D includes input parameters ($t_i \in R^n$) and its corresponding class ($y_i \in Y$).

Formally, the problem can be defined as follows:

$$f^* = \arg \min_{f \in F} \{ \mathcal{L}_D(f(t_i), y_i) \mid \forall (t_i, y_i) \in D \} \quad (1)$$

where F represents the set of all permissible functions, y_i denotes the ground truth (i.e., the expected diagnosis), and $\mathcal{L}_D$ stands for a distance metric indicating the difference between the predicted and actual outcomes, which can be formulated as follows:

$$\mathcal{L}_D = \frac{1}{|D|} \sum_{(t_i, y_i) \in D} V(t_i, y_i) \quad (2)$$

where

$$V(t_i, y_i) = \begin{cases} 0, & f(t_i) = y_i \\ 1, & f(t_i) \neq y_i \end{cases} \quad (3)$$

In other words, we need to find a function f^* that minimizes the cumulative distance $\mathcal{L}_D$ on the dataset D .

However, given that the genetic programming (GP) process yields a numerical outcome in the form of a mathematical formula, certain assumptions must be made to facilitate the implementation of the GP-based classifier. Firstly, the numerical output values must be changed into class values (e.g. 0 and 1). In this manner, GP can find a formula that fits all numeric inputs to the corresponding output values.

As often happens, for a given vector of input features, the provided outcome will hardly match the expected output class. For this reason, the concept of threshold is commonly employed to solve this problem. For a given formula, f^* , the corresponding threshold, T_{f^*} , is determined as the point of maximum separation among the predicted outcomes of f^* evaluated on the entire dataset under consideration. In essence, the threshold value serves to divide the search space into separate regions, each corresponding to a particular class. Consequently, the classification of an input sample is accomplished by determining the region in which its outcome falls.

Typically, the threshold value is chosen in order to maximize some evaluation metrics, such as F-Score or AUC (See Eqs. (9) and (10) below). Note that we used the F-score because of its ability to account for the trade-off between precision and recall, as well as its adaptability to unbalanced datasets. As expressed in Eq. (4), for a given input vector t_i , if the outcome of the formula $f^*(t_i)$ is equal to or greater than T_{f^*} ,

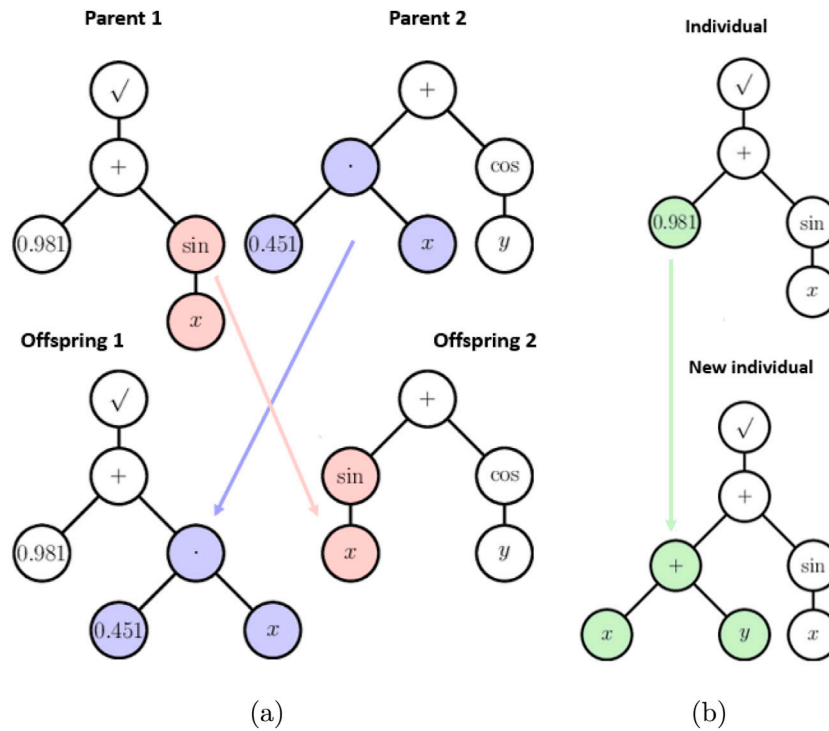


Fig. 2. An instance of (a) crossover and (b) mutation operation.

then the vector is assigned to Class 1; otherwise, it is assigned to Class 0.

$$Class = \begin{cases} 1, f^*(t_i) \geq T_{f^*} \\ 0, f^*(t_i) < T_{f^*} \end{cases} \quad (4)$$

The main motivation behind this approach is to get an interpretable model that, unlike AI black-box approaches, explains the existing relationships between input/output pairs [45,46].

4. Experiments

This Section aims to demonstrate the effectiveness of the GP-based classifier in diagnosing WHF starting from raw bio-clinical data. To accomplish this, we first perform a preliminary study aimed at analyzing the capabilities of the most notable ML-based algorithms in spotting HF. Then, we prove the effectiveness of GP by comparing its results with those derived from the preliminary study. Finally, to explain the correlation between the selected parameters and the given diagnosis, we provide a mathematical formula and its graphical representation.

All the experiments were conducted using a Desktop workstation equipped with a 64-bit Windows 11 Operating System, Intel(R) Core i5-12400 CPU @ 2.50 GHz, 32 GB of RAM, and NVIDIA GeForce RTX 3060 GPU.

4.1. Dataset

The retrospective dataset used in this study includes 519 patients affected by chronic HF in follow-up at University Hospital “San Giovanni di Dio e Ruggi d’Aragona” of Salerno. All procedures followed the Helsinki Declaration and the International Conference of Harmonization/Good Clinical Practice guidelines, and all patients provided informed consent. We created a dataset by analyzing our HF clinic’s registry and collecting the baseline data (first visit) of HF patients who had or did not have WHF in the following 12 months, meeting the following criteria: (i) had an age of at least 18 years; (ii) had an HF diagnosis, according to ESC guidelines regardless of the LVEF [47]; (iii) had been followed for at least 12 months in our

HF ambulatory. Exclusion criteria included: (i) history of malignancy and/or severe renal/hepatic impairment; (ii) echocardiographic images of low quality; (iii) occurrence of acute coronary syndrome within the 12 months previous to the enrollment. Specifically, at the enrollment, we retrospectively collected anamnestic, biochemical, and echocardiographic parameters indicative of WHF or of up-titration of optimal HF medical therapy concerning the previous 12 months. We defined WHF as an episode of exacerbation of HF symptoms and signs with a worsening of the New York Heart Association (NYHA) class and a rising of the echocardiographic values indicative congestion, such as systolic Pulmonary Artery Pressure (sPAP), compared to the previous 12 months’ medical evaluation, needing an augmentation of the dose of loop diuretic, to avoid rehospitalization for HF. All data were recorded on a predefined electronic medical record, including demographics, medical history, clinical and pharmacological treatments, laboratory, electrocardiographic, and echocardiographic data. Prior revascularization through surgical and/or percutaneous procedures, and/or a previous history of myocardial infarction defined the ischaemic etiology of HF. A glomerular filtration rate of less than 60 mL/min per 1.73 m² identified the presence chronic kidney disease (CKD). Details of the enrolled population are depicted in Table 1.

4.2. Echocardiographic measurements

Echocardiographic evaluations and measurements were performed with commercially available equipment (GE Vivid E9) using a 3.5 MHz monoplane ultrasound probe, according to the international American Society of Echocardiography (ASE) and European Association of Cardiovascular Imaging (EACVI) guidelines [48,49]. Two expert operators blinded to clinical data evaluated all parameters offline. LVEF was calculated using the Simpson biplane method, using four and two apical chamber cine-loops. Early-diastolic filling (E) was assessed by positioning the pulsed-wave Doppler sample volume at the tip of the tenting area of the mitral valve in the apical 4-chamber view. Mean e’ was recorded in the basal lateral and inferoseptal LV region in the apical 4-chamber view using Tissue Doppler Imaging. Then, the ratio E/e’ was determined. For the assessment of right ventricular (RV) systolic

Table 1Population baseline characteristics. Data are reported as means $\pm$ SD unless otherwise indicated.

Parameter	Type	Range	Description
Male sex [%]	(yes = 1, no = 0)	77.26	Gender of patients
Age [years]	Numeric	67.69 $\pm$ 12.46	Age of patients
Height [cm]	Numeric	166.14 $\pm$ 8.58	Height of patients
Weight [Kg]	Numeric	78.10 $\pm$ 43.87	Weight of patients
BMI [kg/m ²]	Numeric	28.83 $\pm$ 17.24	Body Mass Index value
BSA [m ²]	Numeric	1.88 $\pm$ 0.30	Body Surface Area value
Weight > 3 kg [%]	(yes = 1, no = 0)	7.32	If weight increased > 3 kg
SBP [mmHg]	Numeric	123.00 $\pm$ 15.00	Systolic Blood Pressure
NYHA	Numeric	2.60 $\pm$ 0.58	New York Heart Association functional class
HR [bpm]	Numeric	75.40 $\pm$ 15.54	Heart Rate
K [mEq/L]	Numeric	4.49 $\pm$ 0.78	Serum Potassium level
Creatinine [mg/dl]	Numeric	1.25 $\pm$ 0.50	Serum Creatinine level
eGFR [mL/min/1.73 m ²]	Numeric	67.72 $\pm$ 27.31	Estimated Glomerular Filtration Rate
CKD [%]	(yes = 1, no = 0)	42.58	Chronic Kidney Disease
Hb [g/dl]	Numeric	12.63 $\pm$ 1.89	Haemoglobin level
EF [%]	Numeric	36.00 $\pm$ 10.00	Ejection Fraction value
HFrEF [%]	(yes = 1, no = 0)	77.85	HF with reduced EF
HFmrEF [%]	(yes = 1, no = 0)	11.17	HF with mid-range EF
HFpEF [%]	(yes = 1, no = 0)	10.98	HF with preserved EF
E/e' ratio	Numeric	11.00 $\pm$ 5.00	Diastolic parameter
Diastolic dysf. [%]	(yes = 1, no = 0)	71.29	Diastolic dysfunction
sPAP [mmHg]	Numeric	37.00 $\pm$ 11.00	Systolic Pulmonary Artery Pressure value
TAPSE [mm]	Numeric	20.00 $\pm$ 4.00	Tricuspid Annulus Plane Systolic Excursion value
Optimized therapy [%]	(yes = 1, no = 0)	81.40	Usage of optimized therapy
CAD [%]	(yes = 1, no = 0)	70.13	Coronary Artery Disease
AF [%]	(yes = 1, no = 0)	32.75	Atrial Fibrillation
DM Type-II [%]	(yes = 1, no = 0)	25.00	Diabetes Mellitus
Diuretic dose titration [%]	(yes = 1, no = 0)	38.43	Usage of Diuretic dose
Diuretic dose $\geq$ 50 mg [%]	(yes = 1, no = 0)	31.40	Diuretic dose $\geq$ 50 mg

function, the tricuspid annulus plane systolic excursion (TAPSE) was obtained by aligning the M-mode linear cursor to the lateral tricuspid annulus in the apical four-chamber view. Right ventricular dysfunction was defined in the case of TAPSE < 17 mm. sPAP was derived from the tricuspid regurgitant jet velocity considering systolic trans-tricuspid pressure gradient, calculated by the modified Bernoulli equation, with the addition of estimated mean right atrial pressure obtained from inferior vena cava dimension/distensibility recorded in subcostal view.

4.3. Data analysis and pre-processing

As summarized in Table 1, the dataset includes 29-dimensional data samples regarding the presence and absence of a WHF. We performed an Exploratory Data Analysis (EDA) [50] in order to accurately study the data distribution in relationship with the considered categories. The analysis counted 519 patients, of whose 281 belonged to category 0 and the remaining 238 belonged to category 1. For the sake of clarity, category 0 refers to patients who never had a WHF while category 1 refers to patients who had WHF. Consequently, TPs (True Positives) refer to patients correctly classified as people who suffered of WHF, while TNs (True Negatives) refer to patients correctly identified as people who never suffered of WHF. Instead, FPs (False Positives) represent patients mistakenly classified as those who never suffered a worsening, while FNs (False Negatives) represent patients mistakenly identified as those who suffered a worsening.

Furthermore, since the EDA highlighted the absence of missing values and duplicate entries, no additional cleaning and normalization steps were performed on the employed dataset. Finally, to mitigate

potential biases, we implemented a 5-fold cross-validation technique in our experiments. This involved partitioning the data into 5 folds. Each fold was used in rotation as the testing set, while the remaining 4 folds served as the training set. This approach results in a training-to-testing ratio of 4:1 for each iteration. By rotating the testing fold across all iterations, we ensured a thorough evaluation of the model's performance across different subsets of the data, thereby enhancing the reliability of our results.

4.4. Evaluation metrics

To assess the classification performance of the utilized models, we evaluated several metrics derived from the confusion matrix for a binary classifier. These metrics include Accuracy (Acc.), Sensitivity (Sens.), Specificity (Spec.), Precision (Prec.), F-Score (F-Mea.), and Area Under the ROC Curve (AUC).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$Sensitivity = \frac{TP}{TP + FN} \quad (6)$$

$$Specificity = \frac{TN}{TN + FP} \quad (7)$$

$$Precision = \frac{TP}{TP + FP} \quad (8)$$

$$F - Score = \frac{2 * Sens * Prec}{Sens + Prec} \quad (9)$$

$$AUC = \frac{Sens + Spec}{2} \quad (10)$$

Table 2

Results achieved by Naive Bayes classifier (NB).

	Acc.	Sens.	Spec.	Prec.	F-Mea.	AUC
Fold 1	0.9327	0.9500	0.9219	0.8837	0.9157	0.9359
Fold 2	0.9231	0.9800	0.8704	0.8750	0.9245	0.9252
Fold 3	0.9231	0.9020	0.9434	0.9388	0.9200	0.9227
Fold 4	0.9423	0.9400	0.9444	0.9400	0.9400	0.9422
Fold 5	0.9806	0.9524	1.0000	1.0000	0.9756	0.9762
Avg.	0.9403	0.9449	0.9360	0.9275	0.9352	0.9404

Table 3

Results achieved by Logistic Regression (LR).

	Acc.	Sens.	Spec.	Prec.	F-Mea.	AUC
Fold 1	0.9519	0.9524	0.9516	0.9302	0.9412	0.9520
Fold 2	0.9038	0.9423	0.8654	0.8750	0.9074	0.9038
Fold 3	0.9327	0.9200	0.9444	0.9388	0.9293	0.9322
Fold 4	0.9423	0.9231	0.9615	0.9600	0.9412	0.9423
Fold 5	0.9709	0.9302	1.0000	1.0000	0.9639	0.9651
Avg.	0.9403	0.9336	0.9446	0.9408	0.9366	0.9391

Table 4

Results achieved by Multi-Layer Perceptron classifier (MLP).

	Acc.	Sens.	Spec.	Prec.	F-Mea.	AUC
Fold 1	0.9135	0.9722	0.8824	0.8140	0.8861	0.9273
Fold 2	0.8750	0.9388	0.8182	0.8214	0.8762	0.8785
Fold 3	0.9519	0.9400	0.9630	0.9592	0.9495	0.9515
Fold 4	0.9231	0.9200	0.9259	0.9200	0.9200	0.9230
Fold 5	0.9223	0.8636	0.9661	0.9500	0.9048	0.9149
Avg.	0.9172	0.9269	0.9111	0.8929	0.9073	0.9190

Table 5

Results achieved by Support Vector Machine (SVM).

	Acc.	Sens.	Spec.	Prec.	F-Mea.	AUC
Fold 1	0.9135	0.9722	0.8824	0.8140	0.8861	0.9273
Fold 2	0.8654	0.9773	0.7833	0.7679	0.8600	0.8803
Fold 3	0.9423	0.9388	0.9455	0.9388	0.9388	0.9421
Fold 4	0.9423	0.9400	0.9444	0.9400	0.9400	0.9422
Fold 5	0.9903	0.9756	1.0000	1.0000	0.9877	0.9878
Avg.	0.9308	0.9608	0.9111	0.8921	0.9225	0.9359

4.5. State-of-the-art results

The definition of WHF in international guidelines is recent [1], and, to our knowledge, only Boshu Ru et al. have addressed the WHF prediction issue in [51]. Nevertheless, they provided a comprehensive overview and comparison of different feature engineering approaches for predicting HF by using a few of classical ML approaches, that is MLP, Logistic Regression, Random Forest, and XGBoost, and they did not use a proper ML-based algorithm capable to predict the occurrence of WHF. Furthermore, as the authors themselves state, their results were not excellent, which does not allow to draw deterministic conclusions. Furthermore, the dataset employed in their study considered anamnestic and biochemical parameters without considering echocardiography variables such as sPAP and therapy, which are critical in managing HF patients.

Basis on this, we explored the most prominent ML-based algorithms for HF detection. However, this first experiment also had the aim of analyzing the existence of correlations between the considered features and the worsening of HF. To this end, we used some famous ML-based approaches, which are also implemented in WEKA framework [52], namely Naive Bayes (NB) classifier, Logistic Regression (LR), Multi-Layer Perceptron classifier (MLP), Support Vector Machine (SVM), J48 Decision Trees (J48), and Random Forest (RF) algorithm. Tables 2, 3, 4, 5, 6, and 7 report the K-Fold results related to each ML-based approach, respectively. Instead, Table 8 compares the average (Avg.) results. As

Table 6

Results achieved by J48 Decision Trees (J48).

	Acc.	Sens.	Spec.	Prec.	F-Mea.	AUC
Fold 1	0.9712	0.9348	1.0000	1.0000	0.9663	0.9674
Fold 2	0.9135	0.9796	0.8545	0.8571	0.9143	0.9171
Fold 3	0.9423	0.9216	0.9623	0.9592	0.9400	0.9419
Fold 4	0.9712	0.9608	0.9811	0.9800	0.9703	0.9710
Fold 5	0.9709	0.9744	0.9688	0.9500	0.9620	0.9716
Avg.	0.9538	0.9542	0.9533	0.9493	0.9506	0.9538

Table 7

Results achieved by Random Forest algorithm (RF).

	Acc.	Sens.	Spec.	Prec.	F-Mea.	AUC
Fold 1	0.9712	0.9545	0.9833	0.9767	0.9655	0.9689
Fold 2	0.9038	0.9792	0.8393	0.8393	0.9038	0.9092
Fold 3	0.9615	0.9592	0.9636	0.9592	0.9592	0.9614
Fold 4	0.9519	0.9245	0.9804	0.9800	0.9515	0.9525
Fold 5	0.9806	0.9750	0.9841	0.9750	0.9750	0.9796
Avg.	0.9538	0.9585	0.9502	0.9460	0.9510	0.9543

Table 8

A summarize of the average results of the most notable ML-based approaches powered by WEKA tool.

	Acc.	Sens.	Spec.	Prec.	F-Mea.	AUC
RF	0.9538	0.9585	0.9502	0.9460	0.9510	0.9543
J48	0.9538	0.9542	0.9533	0.9493	0.9506	0.9538
NB	0.9403	0.9449	0.9360	0.9275	0.9352	0.9404
LR	0.9403	0.9336	0.9446	0.9408	0.9366	0.9391
SMO	0.9308	0.9608	0.9111	0.8921	0.9225	0.9359
MLP	0.9172	0.9269	0.9111	0.8929	0.9073	0.9190

shown in Table 8, the Random Forest (RF) algorithm and J48 Decision Trees (J48) achieved the best results with a 95% average accuracy. In detail, they outperformed NB and LR classifiers by 1%, SVM by 2%, and MLP classifier by 4%, respectively. Furthermore, it is also possible to appreciate the ability of such models to discreetly minimize the number of FPs and FNs. This, in fact, is highlighted by the score obtained from the other evaluation metrics, which range from 89% to 95%.

Such results revealed a valid relationship between the considered features and the WHF.

However, as discussed in Section 2, the considered ML-based approaches do not explain why they generate a particular result (i.e., the given diagnosis) [24]. Consequently, physicians might be reluctant to employ such black-box models, even as simple decision-support tools. Therefore, in order to improve both the results and the model explainability, we investigated the effectiveness of a GP-based classifier in correlating the considered features with the given diagnosis (i.e., the eventual presence of WHF).

4.6. GP-based classifier and results

Referring to Section 3, we implemented the GP-based classifier utilizing a MATLAB® library designed for symbolic data mining, known as GPTIPS (Genetic Programming Toolbox for the Identification of Physical Systems) [53]. It is a variant of the GP approach that employs the Multi-Gene-based Genetic Programming (MGGP) algorithm for formula discovery. In detail, MGGP treats each formula as a single gene within an individual, where an individual is represented by a weighted linear combination of these genes along with a bias term. To determine optimal weight values, MGGP employs the least-squares method to minimize the difference between the expected outcome and the observed output. This individual representation, also referred to as symbolic scaling regression, possesses the unique capability of capturing behavior more effectively than traditional Symbolic Regression (SR) methods.

Table 9

Results achieved by GP-based classifier.

	Acc.	Sens.	Spec.	Prec.	F-Mea.	AUC
Fold 1	0.9808	0.9556	1.0000	1.0000	0.9773	0.9778
Fold 2	0.9519	0.9636	0.9388	0.9464	0.9550	0.9512
Fold 3	0.9615	0.9592	0.9636	0.9592	0.9592	0.9614
Fold 4	0.9615	0.9600	0.9630	0.9600	0.9600	0.9615
Fold 5	0.9806	0.9750	0.9841	0.9750	0.9750	0.9796
Avg.	0.9673	0.9627	0.9699	0.9681	0.9653	0.9663

Table 10

Comparison among the GP classifier and ML-based approaches.

	Acc.	Sens.	Spec.	Prec.	F-Mea.	AUC
GP	0.9673	0.9627	0.9699	0.9681	0.9653	0.9663
RF	0.9538	0.9585	0.9502	0.9460	0.9510	0.9543
J48	0.9538	0.9542	0.9533	0.9493	0.9506	0.9538
NB	0.9403	0.9449	0.9360	0.9275	0.9352	0.9404
LR	0.9403	0.9336	0.9446	0.9408	0.9366	0.9391
SMO	0.9308	0.9608	0.9111	0.8921	0.9225	0.9359
MLP	0.9172	0.9269	0.9111	0.8929	0.9073	0.9190

Therefore, in our experiments, we implemented the GP-classifier by varying different hyper-parameters. For the sake of clarity, only the set-up that provided the best results is shown below:

- *pop_size*: the dimension of considered population = 100;
- *runs*: the number of algorithm iterations = 100;
- *num_gen*: the number of generations at each iteration = 30;
- *tournament_size*: the size of tournament selection criteria = 5;
- *max_depth*: the max depth of solution trees = 7;
- *bias*: the range of bias values [-10; 10];
- *functions*: the considered arithmetic operators [$\times$, $-$, $+$, $/$, $\sqrt{}$, $\exp$].

Table 9 reports the K-Fold results obtained by the GP-based classifier, while Table 10 compares average results with those derived from the preliminary study (i.e., the ML-based approaches provided by WEKA).

As shown, all the metrics achieved a very high score for any fold, then we can confirm the effectiveness of the proposed GP-based classifier in diagnosing WHF (average accuracy close to 97%). Moreover, as it is proven by the other considered evaluation metrics, the proposed classifier better minimizes FPs and FNs. Consequently, as shown in Table 10, the GP-based classifier globally achieved an increase of 1.5% compared to the best ML-based approaches provided by WEKA, i.e. the Random algorithm respectively Forest (RF) and J48 Decisions Trees (J48).

4.7. Formula extraction and explainability

Although the results obtained in the previous experiments confirm the effectiveness of the proposed GP-based classifier, we remark that, unlike the state-of-the-art solutions, the proposal is able to provide a mathematical formula expressing the relationship between the bio-clinical parameters and the given diagnosis. To this end, we derived f^* by running the proposed algorithm on the whole dataset. In detail, we used the same set-up of cross-validation, except for the initial population (randomly chosen).

The outcome is represented by the following equations:

$$f^* = G_1 + G_2 + G_3 \quad (11)$$

$$G_1 = 1.263 \cdot x_{28} + 0.0025 \cdot x_{22}^2 \cdot e^{(-2 \cdot e^{x_{28}})} \quad (12)$$

$$G_2 = -0.00072 \cdot x_{22}^2 \cdot e^{(-2 \cdot x_{25})} \cdot e^{(-2 \cdot e^{x_{28}})} \quad (13)$$

$$G_3 = 5062 \cdot x_{22} \cdot e^{(x_{12} - x_{28})} \cdot e^{(-\exp(\exp(\exp(x_{28}))))} - 0.3014 \quad (14)$$

Table 11

Comparison between the derived formula and K-Fold cross-validation.

	Acc.	Sens.	Spec.	Prec.	F-Mea.	AUC
Formula	0.9672	0.9662	0.9681	0.9622	0.9642	0.9672
K-Fold	0.9673	0.9627	0.9699	0.9681	0.9653	0.9663

where, with reference to Table 1, the input variables x_i represent Creatinine (x_{12}), sPAP (x_{22}), CAD (x_{25}), and Diuretic dose titration (x_{28}), respectively.

Next, as shown in Fig. 3 and with reference to Eq. (4), we used a threshold value equal to 0.35 ($T_{f^*} = 0.35$) which maximize the F-Score. In detail, we can appreciate how our approach mistakenly classified only few observations, obtaining $FP = 9$ and $FN = 8$. Such a result is quite similar to those obtained in the K-Fold cross-validation, as it can be observed by Table 11.

As depicted, the derived formula (f^*) is given by the summation of three functions, that is G_1 , G_2 , and G_3 . While, any G_i is expressed by a combination of non linear terms. Although the interpretation of such expressions is not trivial, several considerations can be performed, which can be used to better understand the relationship existing between considered bio-clinical features and WHF.

Firstly, only four bio-clinical features are included in the formula, that is only these features appear to be decisive for classifying WHF. As shown, they are Creatinine (x_{12}), sPAP (x_{22}), CAD (x_{25}), and Diuretic dose titration (x_{28}).

Furthermore, it should be noted that for $x_{28} = 1$ (i.e., presence of diuretic dose titration), f^* always assumes values greater than the threshold (0.35) for any value assumed by other parameters, which indicates that the patients belonging to this condition are suffering from WHF.

This is a crucial step for understanding the importance of our finding. The increase of the dose of diuretic relates to a congestive profile of the patients with high possibility of WHF occurrence. Therefore, we employed the presence of diuretic dose titration to confirm the effectiveness of the proposed formula in identifying WHF.

On the other hand, when no diuretic is used ($x_{28} = 0$), the outcome of f^* can assume values that are both greater and lower than the threshold. In particular, G_1 depends only on the value assumed by sPAP (x_{22}) through the following:

$$G_1 = 3.4 \cdot 10^{-4} \cdot x_{22}^2 \quad (15)$$

This means that $0.08 \leq G_1 \leq 1.7$, for typical values of x_{22} (from 15 to 71).

Whilst, depending on the value assumed by CAD ($x_{25} \in \{0, 1\}$), G_2 yields the following:

$$G_2 = \begin{cases} -9.74 \cdot 10^{-5} \cdot x_{22}^2, & x_{25} = 0 \\ -1.32 \cdot 10^{-5} \cdot x_{22}^2, & x_{25} = 1 \end{cases} \quad (16)$$

As a consequence, it is $-0.5 \leq G_2 \leq -0.02$ for $x_{25} = 0$ and $-0.066 \leq G_2 \leq -0.003$ for $x_{25} = 1$, respectively.

Finally, we have:

$$G_3 = 0.0013 \cdot e^{x_{12}} - 0.3014 \quad (17)$$

This implies $-0.3 \leq G_3 \leq -0.022$.

As depicted, G_2 is an order of magnitude smaller than G_1 , and therefore has no effect on f^* . Hence, CAD seems does not be relevant for diagnosing WHF. Conversely, the values assumed by Creatinine (x_{12}) can vary the result of f^* .

Ultimately, in absence of diuretic dose titration, Creatinine and sPAP seem to be determinant for diagnosing WHF.

However, in order to give a practical interpretation of these results, we plotted the outcomes of the proposed formula estimated on the whole dataset in a 3d scatter graph, as shown in Fig. 4. As depicted, the graph confirms that Creatinine and sPAP assume a relevant role in the

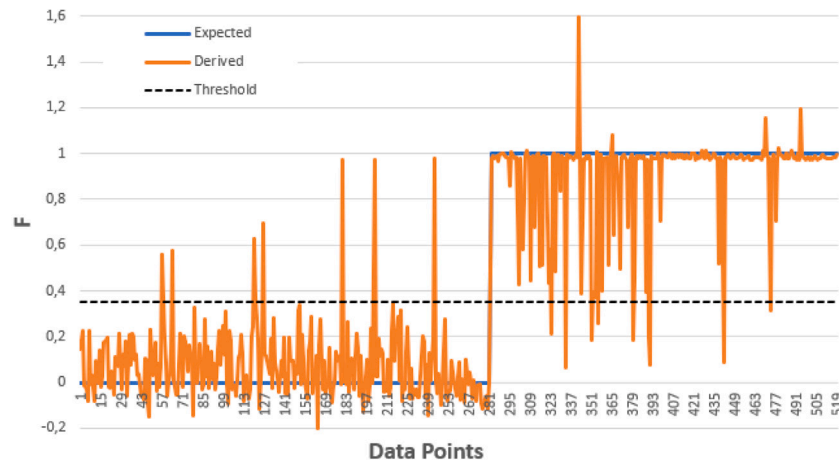


Fig. 3. Predicted vs expected classes. The threshold (0.35) is also depicted.

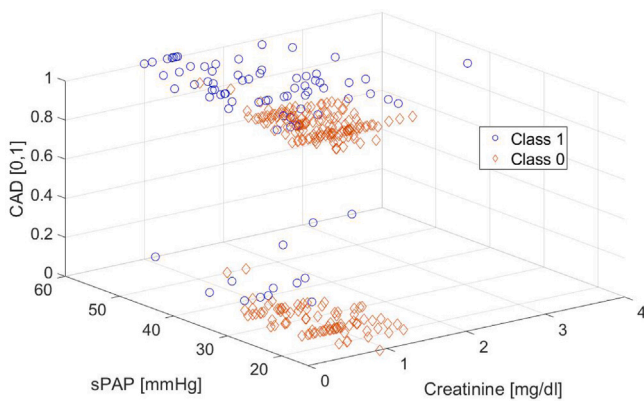


Fig. 4. Graphical representation of the proposed GP-based formula estimated on the whole dataset when no diuretic is used ($x_{28} = 0$).

classifying. Indeed, regardless of the value assumed by CAD, Class 0 is located in a specific area of the graph, roundly for $sPAP \leq 30$ mmHg and $Creatinine \leq 1.5$ mg/dl. Whilst, high values of sPAP or Creatinine could lead to have a WHF.

5. Discussion and limitations

Observing the mathematical model of the proposed classifier, two main deductions can be derived. First, the onset of WHF within 12 months is closely linked to only four variables: creatinine, sPAP, CAD, and diuretics. In particular, diuretics are linked with WHF independently from other parameters. Intuitively, escalating diuretic dosage is much more likely to be a marker of WHF because patients with an escalation in oral diuretics are likelier to have signs/symptoms of congestion. Accordingly, Khan et al. [54] showed that patients with a diuretic dose increase had a significantly higher number of HF hospitalizations (rate ratio, 2.53 [95% CI, 2.10–3.05]), emergency department visits (rate ratio, 1.84 [95% CI, 1.56–2.17]), and home health visits (rate ratio, 1.88 [95% CI, 1.39–2.54]). Indeed, outpatient oral diuretic dose escalation is an important clinical event for HF patients and is linked with increased downstream hospitalizations and healthcare resource utilization [54]. Serum creatinine levels and sPAP play a crucial role in WHF, even in a non-diuretic situation. In detail, low values of creatinine or sPAP seem to prevent WHF. Precisely, in patients with HF, sPAP elevation is one of the hallmark features of HF progression to worsening clinical status, and it is often reversible following appropriate therapy [55]. In this regard, continuous sPAP

monitoring devices, such as CardioMEMS (Abbott Medical, Inc., Abbott Park, IL, USA), are used to detect WHF and prompt actionable responses to pressure elevation [56]. Precisely, CardioMEMS can detect higher cardiac filling pressures, an objective measure of ‘haemodynamic congestion’, estimated to rise more than 2 weeks before the onset of symptomatic clinical congestion [12,57]. However, in this study, we used a non-invasive method without any device, and our algorithm can be easily applied and interpreted. Moreover, CardioMEMS is indicated only for patients with advanced HF (NYHA class III with at least one admission for HF in the past 12 months) [12]. Instead, our algorithm could suggest new indications for these invasive monitoring devices. An increase in serum creatinine, also termed worsening renal function, commonly occurs in patients with HF, especially during acute heart failure episodes [58]; moreover, impaired kidney function is one of the strongest predictors of outcome in HFrEF [58]. Precisely, cardiorenal syndrome includes a spectrum of disorders involving both the heart and kidneys in which acute or chronic dysfunction in one organ may induce acute or chronic dysfunction in the other organ; consequently, it represents the confluence of heart-kidney interactions across several interfaces [59]. Specifically, in our algorithm, an increase in creatinine values is associated with an increase in sPAP which likely excludes over-diuresis and dehydration. Finally, these results show that the proposed GP-based classifier can estimate early WHF and that it is easy to interpret a patient’s condition by inspecting Fig. 4. Consequently, its employment could provide additional benefits for both physicians and patients, such as a more rapid diagnosis, a more targeted cure, and a reduction of hospitalizations.

It is important to acknowledge several limitations to our study, which may restrict the generalizability of our results. First, this is a single-center study and can affect reproducibility within different settings. Indeed, our algorithm should be validated in an external cohort. Moreover, we did not consider biomarkers of increased filling pressure in our dataset, such as B-type natriuretic peptide (BNP) or N-Terminal-Pro (NT-Pro) BNP, which could further enhance the efficacy of the developed algorithm in predicting WHF.

6. Conclusions and future works

The aim of this work has been to enhance diagnostic and prognostic quality in the cardiovascular field, with a particular interest in identifying the elements that can lead to a WHF. Two directions have been followed. Firstly, we have investigated the existence of relationships between bio-clinical parameters and the onset of WHF within 12 months of the first control visit. To this end, we have employed some of the most renowned advanced ML techniques (powered by WEKA) on a real dataset that includes several bio-clinical parameters of patients from

the Cardiology Clinic of the University Hospital of Salerno, Italy. All the ML-based algorithms used have confirmed the existing of correlations between several bio-clinical parameters and WHF. Nevertheless, as is known, most of ML-based models are presented as black boxes, which do not provide any explanation of how the input parameters are related to WHF. This, in many contexts, especially in medicine, is not well accepted, and an interpretable model is desired. To accomplish this, we have proposed a binary classifier based on a GP making usage of a threshold (estimated to obtain the maximum classification accuracy) to turn the numerical results of GP into classes. Such a classifier has shown to achieve better performance compared to the state-of-the-art. Moreover, the proposal is able to provide a mathematical formula expressing the relationship between the identified parameters and the given diagnosis, which leads to an easy interpretation of the derived model. Precisely, the WHF is closely linked to only three bio-clinical parameters, namely Creatinine, sPAP, and CAD. Furthermore, low values of creatinine or sPAP seem to prevent WHF.

The outcomes achieved in our work encourage us to employ the proposed GP-based approach in the future for diagnosing different heart diseases, such as myocardial infarction, aortic aneurysm, angina pectoris, and atherosclerosis. Additionally, we intend to investigate the effectiveness of GP in classifying and predicting Ejection Fraction, Tricuspid Annular Plane Excursion (TAPSE), and Right Ventricular Systolic Function (RVs) values, which represent three highly observed parameters in the cardiovascular field. All of this could offer several benefits for both physicians and patients in terms of faster diagnosis, targeted treatment, and a reduction in hospitalizations.

Funding

This research was funded by the Ministry of Economic Development (MISE), Italy, through the MISE Accordo Innovazione Scienze della vita DM 02/08/2019 under the Project SOLOMAX (SOLOMAX – SocialNetWork of MedicALEXperiences), N. F/260007/05/X51, CUP B49J21031700008 - COR 9257850.

CRediT authorship contribution statement

Valeria Visco: Writing – review & editing, Investigation, Data curation. **Antonio Robustelli:** Writing – original draft, Software, Investigation. **Francesco Loria:** Writing – review & editing, Data curation. **Antonella Rispoli:** Writing – review & editing, Data curation. **Francesca Palmieri:** Data curation. **Alessia Bramanti:** Data curation. **Albino Carrizzo:** Data curation. **Carmine Vecchione:** Supervision. **Francesco Palmieri:** Writing – review & editing, Supervision. **Michele Ciccarelli:** Writing – review & editing, Validation, Supervision, Data curation. **Gianni D'Angelo:** Writing – review & editing, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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